The Particle Theory and States of Matter

- Matter is everything that has mass and takes up space.
- All matter is made up of tiny particles of atoms or molecules.
- The way the particles move and behave depends on the state or phase of the matter.
- The particles in matter are in constant motion. This is known as the Kinetic Theory.
- The three states of matter are solid, liquid and gas.

Particles in a Solid

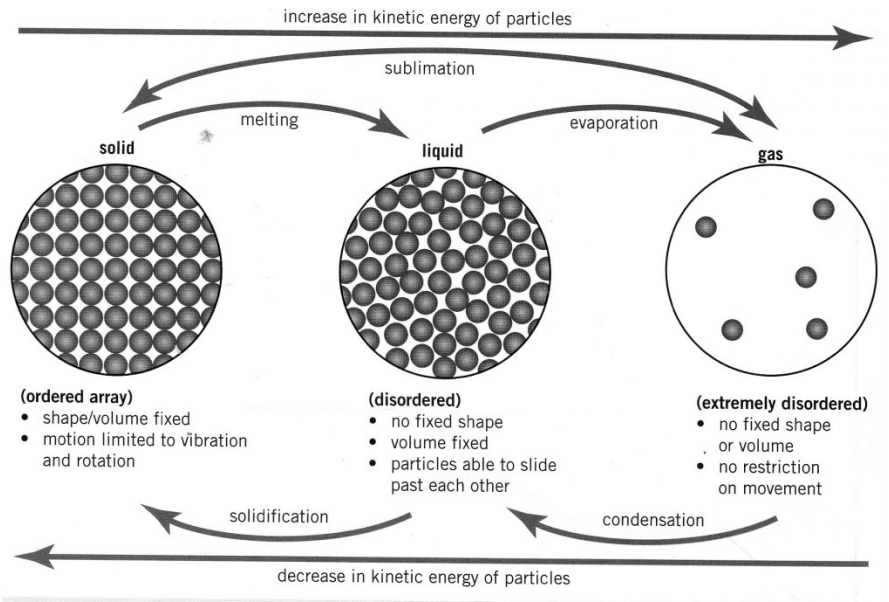
- The particles are packed together tightly, with only small spaces between them.
- The particles are held together by strong forces so they cannot move freely, but vibrate around fixed positions.
- Solids expand when heated and contract when cooled.
- Solids have a fixed shape and volume.
- Solids cannot be compressed.

Particles in a Liquid

- The particles are packed tightly together, with only small spaces between them.
- The particles are held together by weaker forces, but they cannot move freely. They are able to roll over each other. This allows liquids to flow and be poured.
- Liquids expand when heated and contract when cooled.
- Liquids have no fixed shape but have a fixed volume.
- Liquids cannot be compressed.

Particles in a Gas

- The particles in gases are far apart and only held together with very weak forces.
- The particles are able to move freely in all directions.
- Gases have no fixed shape or volume but will fill up the container they are enclosed in.
- Gases will expand a lot when heated and shrink when cooled.
- Gases can become liquids if cooled or compressed enough.



Temperature and Energy

- The temperature of a substance is a measure of the amount of internal energy the substance has.
- A substances state of matter is determined by its internal energy or temperature.
- Heating matter gives its particles more energy so they vibrate faster and move further apart. The matter will expand slightly in volume.
- If enough energy is put into the matter, the particles will have enough energy to overcome the forces holding them together and change phase.
- Cooling matter takes energy away from the particles so they vibrate more slowly and move closer together or shrinks in volume.
- When cooling takes enough energy out of the matter, the particles will move more slowly until they condense and change phase.

Questions

1. Name 2 common substances that are solid at room temperature.

2. Name 2 common substances that are liquid at room temperature.

3. Name 2 common substances that are gas at room temperature.

4. Draw the arrangement of the particles in ice, liquid water and steam.

5. Describe how the particles are moving in each of the states of matter.

6. What is the relationship between particle movement and temperature?

7. Name the phase changes that occur when:
    a. Solid turns to liquid
    b. Gas turns to liquid
    c. Liquid turns to gas
    d. Liquid turns to solid